

Lecture 5

In Class Exercises



Exercise 1

- In HIRE-ASSISTANT, assuming that the candidates are presented in a random order, what is the probability that you hire exactly one time? What is the probability that you hire exactly n times?

Exercise 1 (solution)



- Since HIRE-ASSISTANT always hires candidate 1, it hires exactly once if and only if no candidates other than candidate 1 are hired. This event occurs when candidate 1 is the best candidate of the n , which occurs with probability $1/n$.
- HIRE-ASSISTANT hires n times if each candidate is better than all those who were interviewed (and hired) before. This event occurs precisely when the list of ranks given to the algorithm is $\{1, 2, \dots, n\}$, which occurs with probability $1/n!$.

Exercise 2

- Use indicator random variables to compute the expected value of the sum of n dice.
- Hint: Use 3 random variables: X , Y Z .
- X : The indicator random variable
- Y : The value rolled on a particular die.
- Z : ???

Exercise 2 (solution)

For $i = 1, 2, \dots, 6$, define the indicator random variable

$$\begin{aligned} X_i &= I\{i \text{ is rolled}\} \\ &= \begin{cases} 1 & \text{if } i \text{ is rolled,} \\ 0 & \text{if } i \text{ is not rolled.} \end{cases} \end{aligned}$$

Since each face value has a probability of $1/6$ of being rolled, $E[X_i] = 1/6$, for $i = 1, 2, \dots, 6$.

Exercise 2 (solution)

Now define a random variable Y_j that is equal to the value rolled on die j , where $j = 1, 2, \dots, n$. Then,

$$\begin{aligned} E[Y_j] &= \sum_{i=1}^6 i E[X_i] \\ &= \sum_{i=1}^6 i \frac{1}{6} \\ &= \frac{1}{6} \sum_{i=1}^6 i \\ &= \frac{7}{2}. \end{aligned}$$

Exercise 2 (solution)

Finally, define a random variable Z equal to the sum of the n dice rolls. We want to compute $E[Z]$. Therefore, we have

$$\begin{aligned} E[Z] &= \sum_{j=1}^n E[Y_j] \\ &= \sum_{j=1}^n \frac{7}{2} \\ &= \frac{7n}{2}, \end{aligned}$$

and so $7n/2$ is the expected sum of n dice rolls.